Automated Retail Analytics Pipeline

End-to-End Data Engineering, SQLite Warehousing, and Zero-Touch Reporting

Shisir Bhandari
Rohan Rayamajhi
Sarbesh Timsina
Prajwal Subedi

The Friction & The Fix

Current State (Manual)

Time-Consuming
Daily & monthly reports require hours of manual CSV processing per cycle.

Error-Prone
Human copy-paste workflows introduce inconsistencies in KPI calculations.

Isolated
No centralised data store; impossible to support analytics across regions.

Future State (Automated Pipeline)

Scheduled
100% automated script execution with zero manual intervention.

Validated
Programmatic data cleaning and automated feature engineering.

Centralised
Lightweight SQLite data warehouse feeding directly into interactive dashboards.

System Architecture

Raw CSV
Extraction of unstructured transactional data.

Python ETL
Cleaning, deduplication, and transformation.

SQLite Warehouse
Structured, serverless star-schema storage.

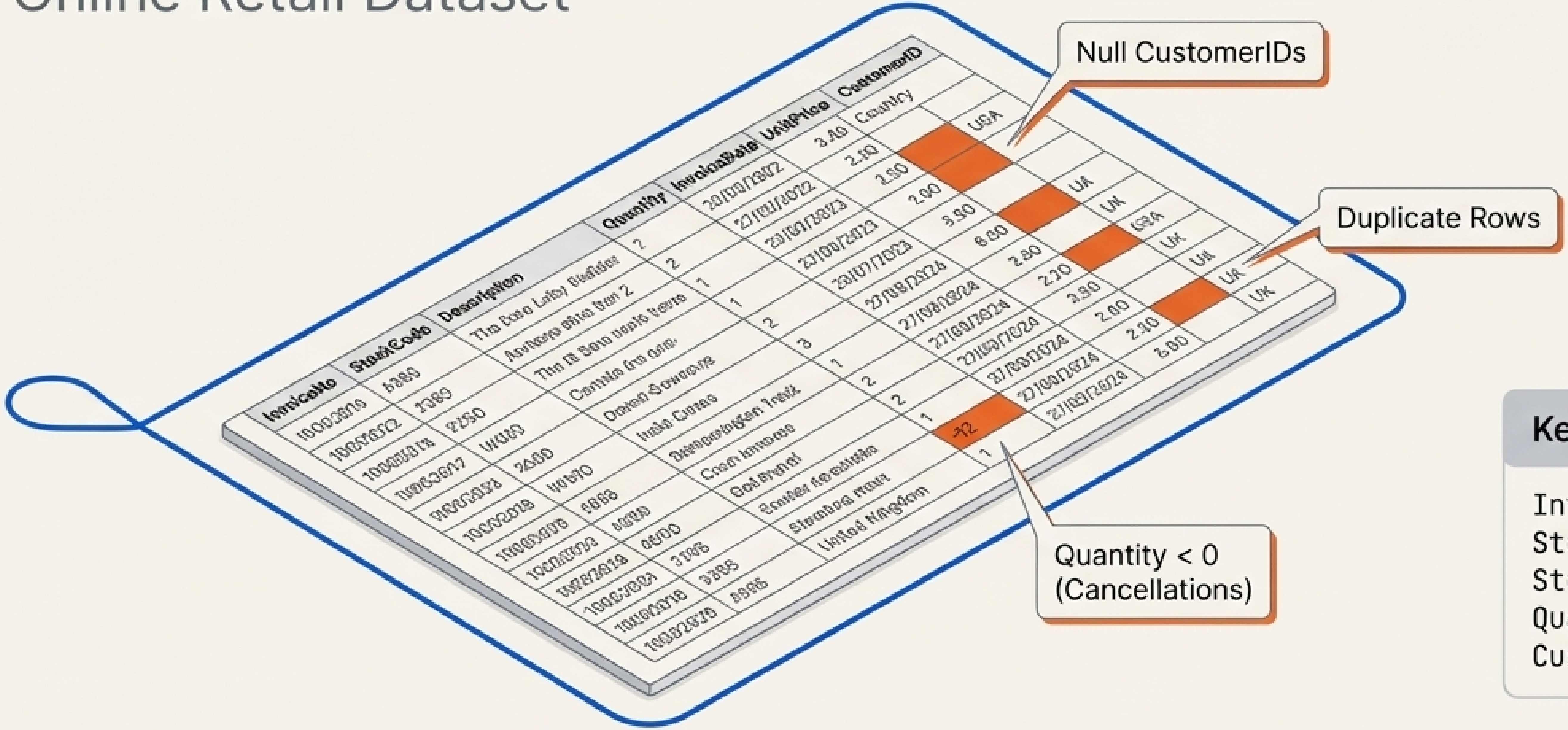
Analytics & KPIs
Programmatic generation of reports and metrics.

Streamlit Dashboard
Interactive visualisation layer.

A simulated cron-based scheduler automates this entire sequence at daily intervals.

The Raw Material

Online Retail Dataset



Key Extracted Columns:

```
InvoiceNo & InvoiceDate
StockCode & Description
StockCode & Description
Quantity & UnitPrice
CustomerID & Country
```

Data Profile: Transaction-level sales records requiring extensive standardisation.

The Refinery: Python ETL Process

1. Extract

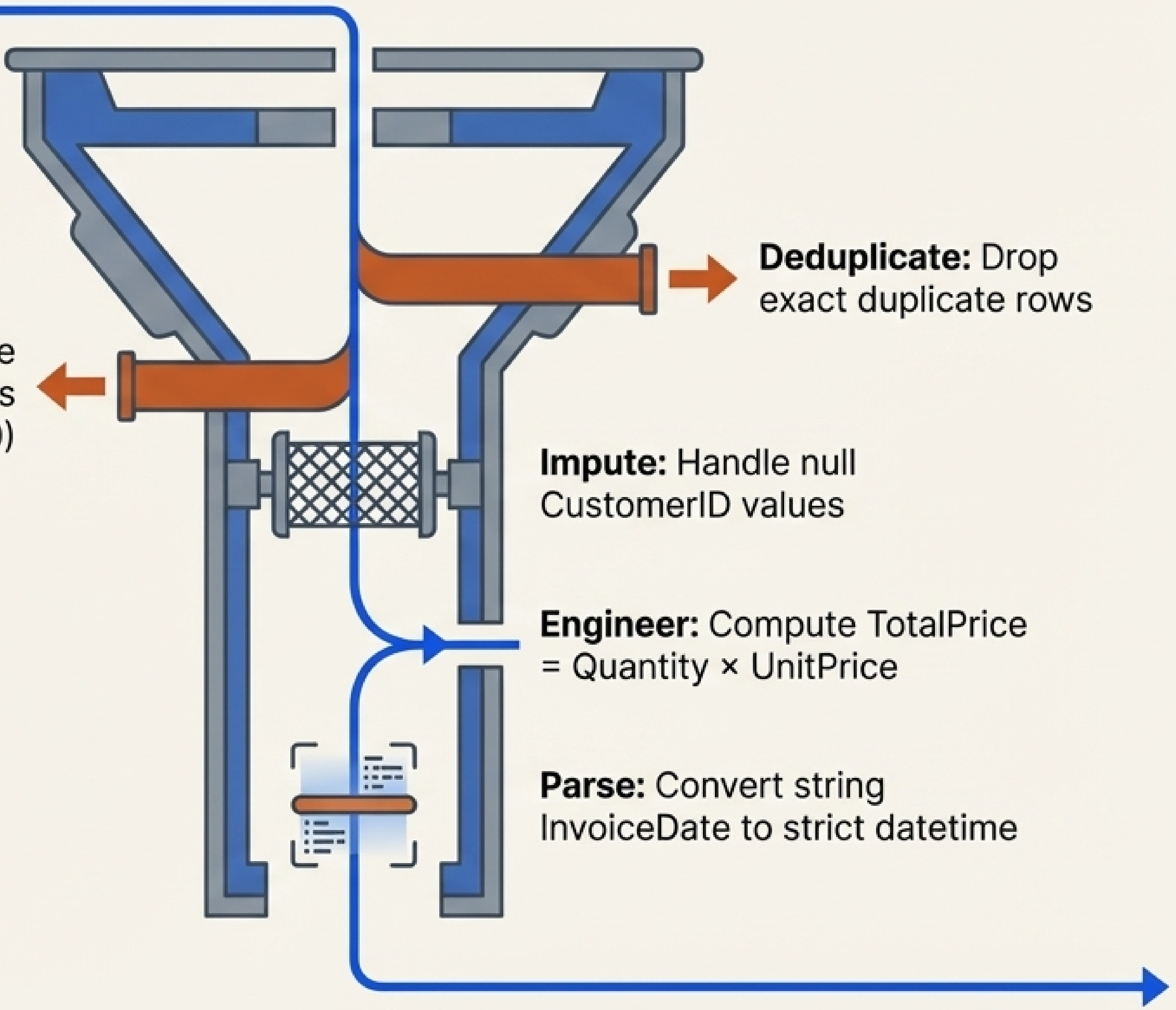
Ingest CSV using Pandas
with schema validation.

2. Transform

The Filtration Engine

3. Load

Stage clean DataFrame for
database insertion.



PP Neue Montreal

The Vault: Storage Design

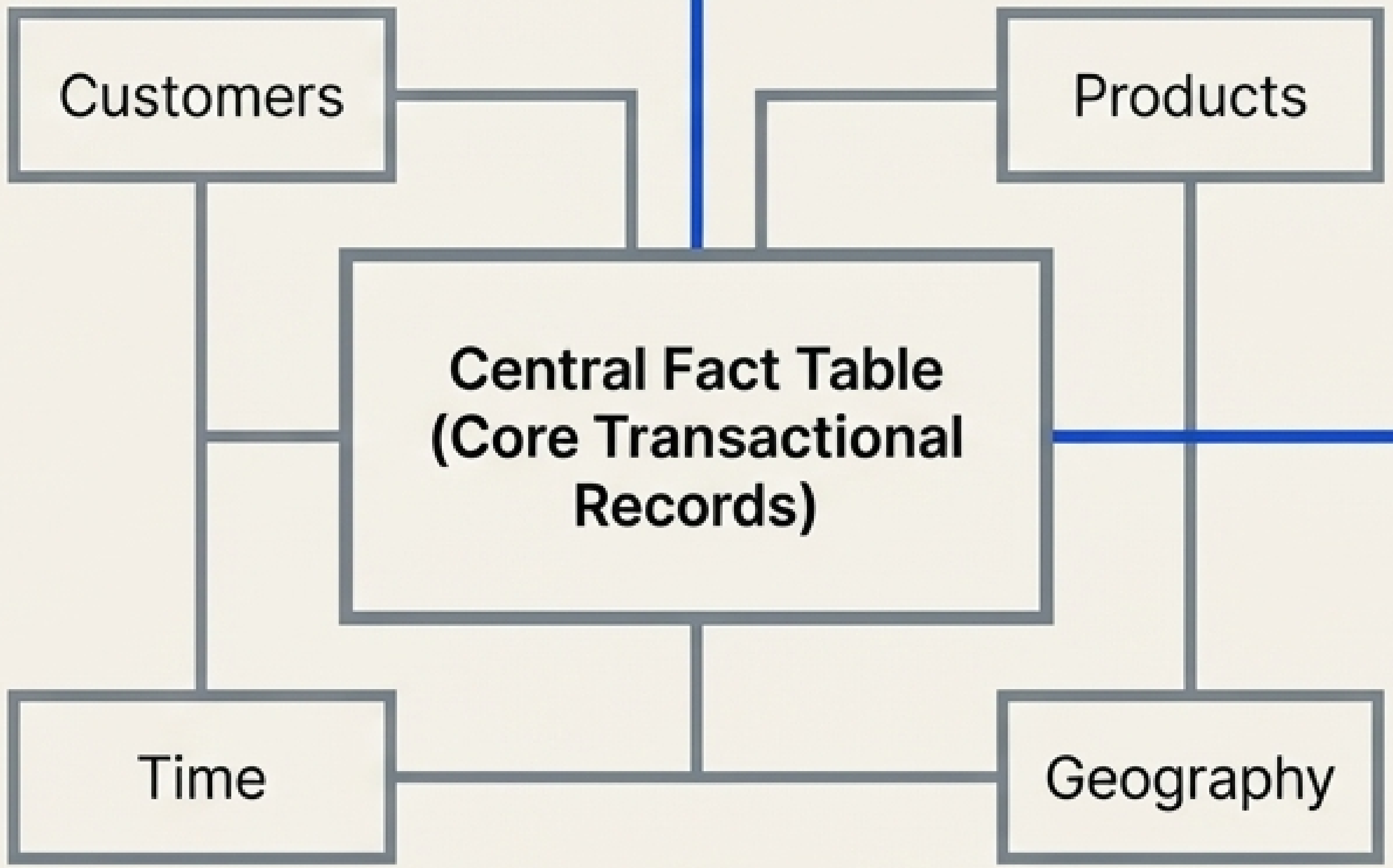
Engineering Choice: SQLite

Lightweight & Serverless

Requires no dedicated server infrastructure, making it highly portable.

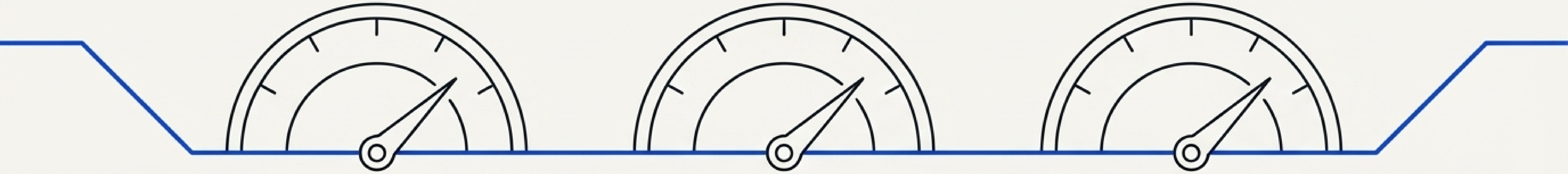
Seamless Integration

Natively bridges with Pandas and Python analytics scripts.



Optimised for rapid query performance during KPI generation and daily reporting.

The Analytical Engine: RFM Segmentation



Recency (R)	Frequency (F)	Monetary (M)
Days since last purchase. (e.g., 7 days = High R score)	Total discrete purchases. (e.g., 12 orders/90 days = High F score)	Total lifetime spend. (e.g., £4,200 = High M score)

Generated Behavioural Segments:

- Champions
- Loyal
- At Risk
- Lost
- New Customers

The Heartbeat: Scheduled Automation

Mechanism

Simulated cron-based automation using the Python schedule library.

Impact

Transforms a reactive manual chore into a proactive, always-on data engine.

```
schedule.every().day.at('02:00')
  run_pipeline()
  run_etl() → calculate_rfm() → generate_reports()
```

The Control Room: Logging & Monitoring

Infrastructure: Python logging module routing to structured files.

logs/scheduler.log

```
2025-11-01 02:00:01 | INFO | Scheduler started
2025-11-01 02:00:02 | INFO | Starting ETL Pipeline
2025-11-01 02:00:15 | INFO | RFM calculation completed successfully
2025-11-01 02:00:22 | INFO | Pipeline completed successfully
```

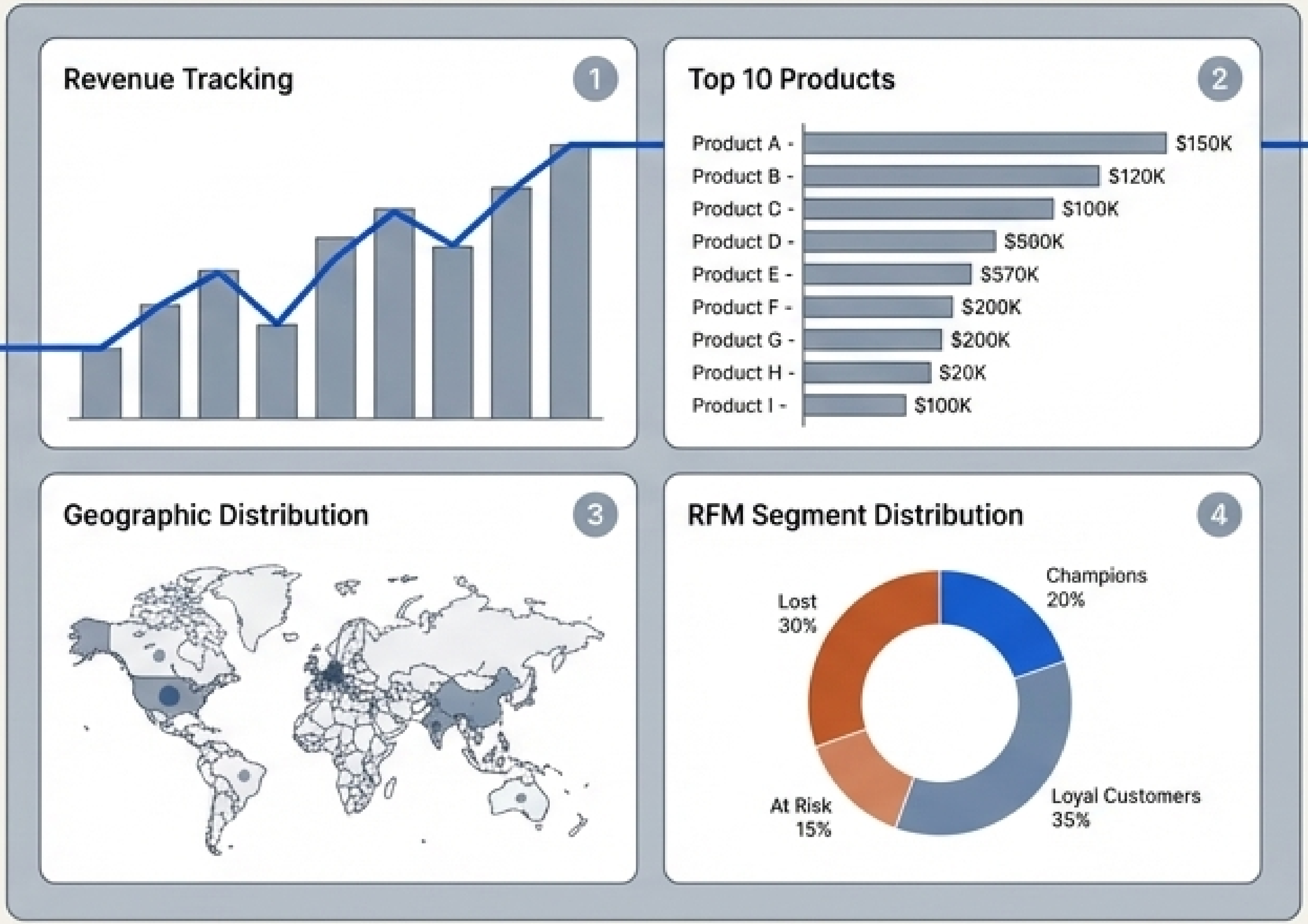
Technical Fix: Implemented os.makedirs() prior to logging configuration, resolving absolute paths to prevent misrouted log files and ensure total pipeline transparency.

The Output: Streamlit Dashboard

Deliverables: Automated KPI generation and visual report distribution.

Key Visualised Metrics:

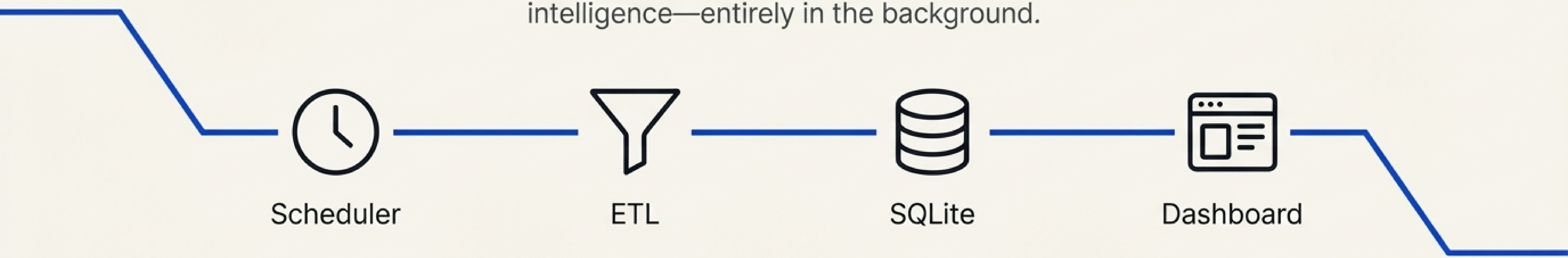
1. **Revenue Tracking:** Daily time-series trends and aggregated Month-over-Month growth.
2. **Product Performance:** Top 10 products by total revenue generation.
3. **Geographic Distribution:** Sales breakdowns by country and region.
4. **Audience Insights:** Repeat customer loyalty analysis and full RFM segment distribution charts.



Synthesis: Zero-Touch Analytics

Running While You Sleep

The Achievement: An end-to-end architecture that extracts raw data, computes complex behavioural segmentations, and visualises business intelligence—entirely in the background.



100%

Automated Execution

5

Live KPI Reports Generated Daily

0

Manual Steps Required

The Horizon: Future Roadmap

Phase 1: Orchestration

Integration with Apache Airflow for DAG-based management and automated retry/alerting logic.

Phase 2: Cloud Deployment

Migration to AWS S3 / GCS data lakes and upgrading SQLite to cloud-hosted PostgreSQL.

Phase 3: Real-Time Streaming

Implementing Kafka for live data ingestion and real-time Streamlit dashboard refreshes.

Phase 4: Machine Learning

Deploying churn prediction models, demand forecasting, and personalised customer recommendations based on historical RFM data.